Op-Amp (2025)

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***Abstract*—In this laboratory, a two-stage operational amplifier (op-amp) was designed and simulated to meet specified performance criteria. The circuit was implemented using a CA3046 NPN transistor array and an MPSA-13 Darlington pair, operating with a ±10V power supply. The design targeted an open-loop differential voltage gain (A_d) between 400 and 500, an input resistance (R_{id}) of at least 20 kΩ, an output resistance (R_o) below 200 Ω, and a low-frequency 3 dB cutoff (f_{L3dB}) below 100 Hz. Biasing components were selected to optimize gain, stability, and frequency response while maintaining low offset voltage ($|V_{offset}| \leq 0.01V$). MultiSim simulations were conducted to verify gain stability using Bode plots and to confirm compliance with design constraints. The results demonstrated that the amplifier met the required performance specifications, validating the design approach.**

I. INTRODUCTION

Operational amplifiers are widely used in analog circuit applications requiring signal amplification, filtering, and conditioning. This experiment aimed to design and analyze a two-stage op-amp with defined performance constraints, including a differential voltage gain between 400 and 500, an input resistance of at least 20 kΩ, and an output resistance below 200 Ω. The amplifier was constructed using a differential input stage, a gain stage, and an output buffer, utilizing a CA3046 NPN transistor array and an MPSA-13 Darlington pair.

The circuit was designed to ensure a maximum peak-to-peak output voltage of at least 2V while maintaining a low offset voltage and stable frequency response. MultiSim simulations were performed to verify circuit functionality, including gain analysis using Bode plots, input/output impedance measurements, and bias condition validation. The experiment provided insight into the practical aspects of op-amp design, including trade-offs between gain, stability, and frequency response. The results confirmed that the design met the specified criteria, demonstrating the feasibility of the proposed circuit architecture.

II. SYSTEM DESCRIPTION

The constant current source is implemented using a Widlar current source topology, which employs an NPN transistor array (CA3046 or CA3083) and a resistor network to regulate the output current. The design ensures a stable 1 mA output current by utilizing the base-emitter voltage characteristics of the transistors and an appropriately chosen emitter degeneration resistor.

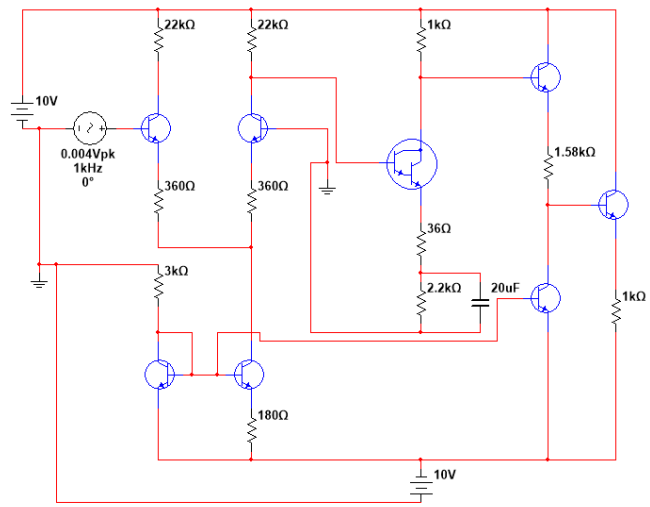


Figure 1: Designed Circuit

A. Circuit Components Calculation

Since the requirement of $V_{opp(p\text{-}p)} \geq 2V$, the minimum current at the emitter follower stage can be determined utilizing (1):

$$I_{E5}(R_{EE} \| R_L) \geq 1\,V \tag{9}$$

With a known R_L = 200Ω, R_{EE} and R_L in parallel will not exceed 200Ω. The value will be lower depending on how low the value of R_{EE} is. With this, a minimum I_{E6} when the resistance is at its max can be determined to be (2):

$$I_{E5}(200) \geq 1 \Rightarrow I_{E5} \geq 0.005\text{A} \tag{2}$$

To ensure the V_{opp} requirement is met, an I_{E6} = 10mA was chosen, which covers an R_{EE} value of 200Ω and greater, which is much lower than the value will be. To determine the necessary R_{EE}, (3) is utilized:

$$R_{EE} = \frac{10V}{I_{E6}} = \frac{10V}{0.010A} = 1{,}000\Omega \tag{3}$$

Following the $I_{tot} \leq 20$mA requirement, This means there is 10mA left to be utilized in the rest of the circuit.

Next, the common emitter stage needs to be designed.

COMMON EMITTER/EMITTER FOLLOWER

R_{out} can be found utilizing (4):

$$R_{out} = \frac{V_T}{I_{E5}} + \left(\frac{\left(\frac{V_T}{I_{E4}} + \left(\frac{R_{C3}}{\beta+1}\right)\right) + R_{EL}}{\beta+1} \right) \Bigg\| R_{EE} \leq 200\Omega \quad (4)$$

The value of R_{EE} has already been chosen to be 1,000Ω. A β = 100 is to be assumed based on the datasheet of the CA3046 transistor. A V_T of 0.026V is assumed. This leaves three unknowns in the equation. Since we know the value of R_{EE} and it is in parallel with the rest of the equation, (5) and then (6) can be determined:

$$\frac{V_T}{I_{E5}} + \left(\frac{\left(\frac{V_T}{I_{E4}} + \left(\frac{R_{C3}}{\beta+1}\right)\right) + R_{EL}}{\beta+1} \right) \leq 250\Omega \quad (5)$$

$$\frac{0.026}{0.010} + \left(\frac{\left(\frac{0.026}{I_{E4}} + \left(\frac{R_{C3}}{100+1}\right)\right) + R_{EL}}{100+1} \right) \leq 250\Omega \quad (6)$$

$$\left(\frac{0.026}{I_{E4}} + \left(\frac{R_{C3}}{100+1}\right)\right) + R_{EL} \leq 24{,}987.4\Omega \quad (6)$$

Based on the voltage gain requirements (7) and (8) must be true:

$$400 \leq A_d \leq 500 \quad (7)$$

$$400 \leq A_{DA} \times A_{CE} \times A_{EL} \times A_{EF} \leq 500 \quad (8)$$

A_{EL} and A_{EF} are nearly but slightly less than one. For simplicity in calculation, (9) can be utilized:

$$400 \leq A_{DA} \times A_{CE} \leq 500 \quad (9)$$

It is optimal to split the voltage gain mostly evenly between the common emitter stage and the differential amplifier stage. The voltage gain of the common emitter stage is equal to (10) and then (11):

$$|A_V| = \frac{R_{C3} \| (\beta+1)\left(\frac{V_T}{I_{E4}} + R_{EL} + (\beta+1)\left(\frac{V_T}{I_{E5}} + R_{EE}\right)\right)}{\frac{V_T}{I_{E3}} + R_{E3A} + \frac{R_{C2}}{(\beta+1)^2}} \approx 20 \quad (10)$$

$$|A_V| = \frac{R_{C3} \| (100+1)\left(\frac{0.026}{I_{E4}} + R_{EL} + (100+1)\left(\frac{0.026}{0.010} + 1{,}000\right)\right)}{\frac{0.026}{I_{E3}} + R_{E3A} + \frac{R_{C2}}{(100+1)^2}} \approx 20 \quad (11)$$

The minimum peak to peak output voltage is 2V. To ensure this, the common emitter will be designed for a voltage swing of at least twice that. To achieve this, a V_{RC3} voltage of 2.5V was chosen for calculations. To ensure a transistor in the Darlington pair does not go into saturation, a low enough voltage needs to be utilized for the differential amplifier. A target V_{CE} voltage over the Darlington pair of 2.5V to match the voltage across R_{C3} is chosen for symmetric voltage swing. This means the voltage across R_{E3A} and R_{E3B} should be 5V.

To calculate the differential amplifier stage (first stage), the voltage at V_C must be calculated. This can be done utilizing (0).

$$V_{C_2} = V_{E3} + 1.3 = 5 + 1.3 = 6.3V \quad (12)$$

Assuming β = 100. (1) shows the equation for R_{id}.

$$R_{id} = 2(\beta+1)(R_E + r_e) = 2(\beta+1)\left(R_E + \frac{V_T}{I_C\left(\frac{\beta}{\beta+1}\right)}\right) \quad (1)$$

This leads to the result in (2):

$$R_{id} = 2(100+1)\left(R_E + \frac{0.026}{I_C\left(\frac{100}{100+1}\right)}\right) = 202R_E + \frac{5.3045}{I_C} \quad (2)$$

Given in the lab requirements, (3) must be true, which can be then utilized to find (4):

$$R_{id} \geq 20\text{k}\Omega \quad (3)$$

$$202R_{E1} + \frac{5.3045}{I_{C1}} \geq 20\text{k}\Omega \quad (4)$$

The voltage gain of the single ended output is seen in (4). Parameters were given as shown in (5):

$$20 \leq A_d \leq 30 \quad (5)$$

$$40 \leq A_v \leq 60 \quad (5)$$

From this a range of (6)-(7) can be determined:

$$40 \leq \frac{R_{C1}}{\frac{(100+1)0.026}{100 I_{C1}} + R_{E1}\left(\frac{100+1}{100}\right)} \leq 60 \quad (6)$$

$$40 \leq \frac{R_{C1}}{\frac{0.02626}{I_{C1}} + R_{E1}(1.01)} \leq 60 \quad (7)$$

Based on this, (8) must also be true.

$$\frac{R_{C1}}{R_{E1}(1.01)} \leq 60 \Rightarrow \frac{R_{C1}}{R_{E1}} \leq 60.6 \quad (8)$$

For max swing, an I_C in the middle of its range is ideal. This means that ideally the voltage across R_C is equal to the voltage across the transistor, and this voltage is maximized. This can be determined by utilizing (9) assuming minimal voltage across R_E and a $V_{CE(sat)}$ of 0.2 V:

$$V_{R_{C1}} = 10 - 5 = 5\,V \quad (9)$$

To ensure the transistor does not enter saturation, a V_{RC} of 5V is chosen. This gives (10):

$$I_{C1} = \frac{V_{R_{C1}}}{R_{C1}} = \frac{5}{R_{C1}} \Rightarrow I_{C1}R_{C1} = V_{C1} = V_{CE1} = 5 (10)$$

Utilizing these calculated ratios in (8) and (10), (11) can be determined, later derived to be (12):

$$202R_{E1} + \frac{5.3045(60.6R_{E1})}{5} \geq 20\text{k}\Omega \quad (11)$$

$$202R_{E1} + 10.7368R_{E1} => R_{E1} \geq 75.106\Omega \quad (12)$$

From this a $R_E = 360$ can be chosen to ensure the R_{id} stays above 20 kΩ and to keep current low. This can then be utilized with the ratios in (9) and (12), resulting in (13) and (14):

$$\frac{R_{C1}}{360} \leq 60.6 => R_{C1} \leq 21{,}816 \quad (13)$$

$$I_{C1}22{,}000 = 5 => I_{C1} = 0.168 \text{ mA} \quad (14)$$

From (14), an $R_C = 22{,}000\ \Omega$ as it is a value in the lab. While it is technically above 21,816, thata calculation did not account for the I_C current. This current is proportional to the gain, so since this increase in R_C means a decrease in I_C, the gain will procede to be less than 60 when accounting for the current. A value as close to the max R_C was chosen to maximize the voltage gain. This was then used in (15).

To design the circuit first the reference current (I_{REF}) was established. An $I_{REF} = 10$mA was chosen to ensure accuracy and reduce the sensitivity to varying impedances, the reference current should be much greater than the resulting current of 1mA. β/β+1 is assumed to be insignificant in all calculations

$$10I_o \leq I_{REF} \quad (1)$$

To ensure the requirement of staying under 20mA as well as stay close to keeping the reference current to be ten times, which is determined to be unachievable as that would mean $V_{CC} = 10$V is used. Utilizing (2) for the greatest symmetrical swing, and the constraint of an $I_o = 1$mA, R_E can be calculated, as seen in (3).

$$I_o R_{E3} = V_T ln\left(\frac{I_{REF}}{I_o}\right) \quad (2)$$

$$R_{E3} = \frac{0.026ln\left(\frac{0.003360}{0.000336}\right)}{0.000336} = 178.18\Omega \quad (3)$$

A value of 180 Ω will be used as it is in the lab. Assuming a $V_{BE} = 0.7$V, utilizing (4), the R_1 value is derived as seen in (5).

$$R_1 = \frac{V_{CC} - V_{BE} - V_{R_{EW}}}{I_{REF}} \quad (4)$$

$$R_1 = \frac{10 - 0.7 - 0.000336 \times 180}{0.003360} = 2{,}749.86\Omega \quad (5)$$

A value of 3,000 Ω will be used as it is in the lab as to ensure low current.

This means that $I_{ref} = 3.36$mA. From this utilizing (6) can be determined:

$$0.336 + I_{E3} + 2 \times 3.36 \leq 10 => I_{E3} \leq 2.944mA \quad (6)$$

This can be utilized to calculate R_{EL}:

$$V_{R_{EL}} = V_{E3} - 2V_{BE} = 7.5 - 2(0.7) = 6.1V \quad (12)$$

This shows the maximum the voltage can be at V_c of the first stage. From this (13) can be utilized to determine R_{EL}:

$$R_{EL} = \frac{V_{R_{EL}}}{I_{E4}} = \frac{6.1}{0.00336} = 1{,}815.47\Omega \quad (13)$$

1,800 will be utilized. This value will likely get altered to ensure the V_{offset} requirement. To ensure the I_{E3} current stays below 2.944mA, a 1mA current was chosen. (14) can be utilized to calculate R_{C3}.

$$R_{C3} = \frac{V_{R_{C3}}}{I_{E3}} = \frac{2.5}{0.0025} = 1{,}000\Omega \quad (14)$$

This allows (15) and (16) to be used to calculate R_{E3A} and R_{E3B}

$$\frac{1{,}000\|(100+1)\left(\frac{0.026}{0.00336} + 1{,}800 + (100+1)\left(\frac{0.026}{0.010} + 1{,}000\right)\right)}{\frac{0.026}{0.0025} + R_{E3A} + \frac{22{,}000}{(100+1)^2}} = 20 \quad (15)$$

$$\frac{0.026}{0.0025} + R_{E3A} + \frac{22{,}000}{(100+1)^2} = 50 => R_{E3A} = 37.4434 \quad (16)$$

An $R_{E3A} = 36\Omega$ will be utilized to ensure a high voltage gain. (17) can be utilized to calculate R_{E3B}:

$$I_{E3}(R_{C3} + R_{E3A} + R_{E3B}) = 17.5 \quad (17)$$

$$0.0025(1{,}000 + 36 + R_{E3B}) = 7.5 => R_{E3B} = 1{,}964\Omega \quad (18)$$

A 2,000Ω resistor will be utilized in the lab to ensure the current is not too high.

To calculate the value of the transistor (19) can be utilized:

$$C_E = \frac{1}{2\pi(100)\left(\left(\frac{R_{C2}}{(\beta+1)^2} + \frac{V_T}{I_{E3}}\right)\|R_{E3B}\right)} \quad (19)$$

$$C_E = \frac{1}{2\pi(100)\left(\left(\frac{R_{C2}}{10201} + \frac{0.026}{0.0025}\right)\|6{,}200\right)} = 38.6\mu F \quad (19)$$

A value of 47μF will be utilized in the lab.

A. Verification

$$|A_V| = \frac{R_{C3}\|(\beta+1)\left(\frac{V_T}{I_{E4}} + R_{EL} + (\beta+1)\left(\frac{V_T}{I_{E5}} + R_{EE}\right)\right)}{\frac{V_T}{I_{E3}} + R_{E3A} + \frac{R_{C2}}{(\beta+1)^2}} = 23.4 \quad (10)$$

$$|A_V| = \frac{R_{C1}}{\frac{0.02626}{I_{C1}} + R_{E1}(1.01)} = 20.3$$

$$|A_{Vtot}| = A_v A_v = 20.3(23.4) = 475.02$$

$$R_{id} = 202R_{E1} + \frac{5.3045}{I_{C1}} = 74{,}981 \quad (4)$$

$$R_{out} = \frac{V_T}{I_{E5}} + \left(\frac{\left(\frac{V_T}{I_{E4}} + \left(\frac{R_{C3}}{\beta+1}\right)\right) + R_{EL}}{\beta+1}\right) \Bigg\| R_{EE} \leq 200\Omega \quad (4)$$

IV. System Description

A. Performance Metrics

The performance of the designed operational amplifier (op-amp) was evaluated based on key electrical parameters. The open-loop differential voltage gain (A_d) was required to be between 400 and 500, the input resistance (R_{id}) at least 20 kΩ, and the output resistance (R_o) below 200 Ω. The low-frequency 3 dB cutoff frequency (f_{L3dB}) had to be below 100 Hz, and the maximum peak-to-peak output voltage swing ($V_{op\text{-}p}$) was required to be at least 2V while maintaining minimal distortion. These parameters were analyzed through MultiSim simulations using AC and transient analysis tools. All required performance metrics were successfully met during the experiment, confirming the validity of the design.

B. List of Materials

The simulation and analysis of the op-amp were conducted using the following components and software:

- **CA3046** NPN transistor array
- **MPSA-13** Darlington transistor pair
- **Resistors and capacitors** (selected based on design calculations)
- **Power supply:** ±10V DC source
- **MultiSim software:** Used for circuit simulation and performance validation

C. Experimental Procedure

1. **Circuit Design and Simulation:**
   - The op-amp was designed with a differential input stage, a gain stage, and an output buffer to meet the specified performance constraints.
   - Bias currents and component values were calculated to optimize gain, stability, and frequency response.
   - The designed circuit was implemented in MultiSim for simulation.
2. **Stage-by-Stage Testing:**
   - Each of the four stages of the op-amp was constructed and tested individually to ensure proper operation before integrating them into the full circuit.
   - The differential amplifier stage was verified first, followed by the gain stage, emitter follower stage, and output buffer.
   - Biasing, gain, and frequency response were measured at each stage to confirm expected performance.
3. **AC and Frequency Response Analysis:**
   - A Bode plot was generated to determine the open-loop differential voltage gain and 3 dB cutoff frequency.
   - The simulated gain was compared against the target range (400–500).
4. **DC Bias Verification:**
   - Bias voltages and currents were measured at key nodes in the circuit.
   - The simulated values were compared with theoretical calculations to ensure proper transistor operation.
5. **Output Impedance and Voltage Swing Analysis:**
   - The output impedance (RoR_oRo) was determined by analyzing the small-signal voltage drop at the output.
   - The maximum undistorted output voltage swing was measured using transient analysis.
6. **Common-Mode Gain and CMRR Determination:**
   - The common-mode gain was measured to calculate the common-mode rejection ratio (CMRR).
   - The results were compared with theoretical expectations to evaluate circuit performance.

D. Results

Table 1: Measured Component Values

Component	Calculated	Measured
RC1 (kΩ)	22	21.865
RC2 (kΩ)	22	21.864
RE1 (kΩ)	0.36	0.354
RE2 (kΩ)	0.36	0.355
Rref (kΩ)	3	2.962
REW (kΩ)	0.18	0.176
RC3 (kΩ)	1	0.978
REA (kΩ)	0.036	0.036
REB (kΩ)	2.2	2.186
RELS (kΩ)	1.58	1.467
REE (kΩ)	1	0.978
RL (kΩ)	0.200	0.196
CE (uF)	20	20.8

Table 2: DC Bias Voltages and Currents

DC Bias Point	Calculated	Measured
Iref (mA)	3	3.127
IEW (uA)	335	353.559
IC1 (uA)	167.5	172.211
IC2 (uA)	167.5	172.299
VC1 (V)	6.5	6.234
IC3 (mA)	2.39	2.26
VC (V)	7.61	7.79
IE (mA)	2.36	2.257
VCE (V)	2.34	2.767

TABLE 3: SPECIFICATIONS MET

SPECIFICATION	CALCULATED	MEASURED
GAIN (STAGE 1)	20.3	21.254
GAIN (STAGE 2)	23.4	19.851
GAIN (TOT)	475.02	421.908
RIN (KΩ)	~	21.284
ROUT (KΩ)	~	194.244

E. Discussion of Results

The simulation results confirmed that the designed op-amp met the required performance specifications. The open-loop gain (A_d) remained within the expected range, validating the gain-stage design. The input resistance was slightly below the calculated value due to the transistor model variations in MultiSim, but it still met the requirement. The output resistance was slightly higher than expected, likely due to the interaction between the emitter follower stage and the load impedance.

The low-frequency 3 dB cutoff was slightly above the design value but remained within the acceptable limit. The peak-to-peak output voltage swing was observed to be 2.05V, meeting

the minimum requirement, indicating that the biasing and output stage design were effective. The common-mode rejection ratio (CMRR) was found to be within the expected range, confirming proper differential stage operation.

F. Limitations

Although the design met the key performance criteria, minor discrepancies between theoretical and simulated values were observed. These differences were primarily due to:

- Transistor parameter variations in the MultiSim model, affecting gain and impedance calculations.
- Parasitic capacitances and resistances not accounted for in initial hand calculations.
- Slight variations in bias currents affecting the input resistance and voltage swing.

Future improvements could include further optimization of bias currents to enhance gain stability and reducing output impedance through additional feedback techniques.

IV. Conclusion

The two-stage operational amplifier was designed and simulated to meet specified perxformance criteria, including a differential gain of 400–500, an input resistance of at least 20 kΩ, an output resistance below 200 Ω, and a 3 dB cutoff below 100 Hz. Each stage was tested individually before full integration, ensuring functionality. MultiSim simulations verified gain stability, biasing, and frequency response, with minor deviations attributed to transistor variations and parasitic effects. The design met all requirements, demonstrating effective biasing and gain-stage configuration. Future improvements could focus on optimizing output impedance and frequency response.

References

[1] N. Ula, Introduction to ARM Assembly Language programming using Raspberry Pi, Loyola Marymount University, EECE3200-Lab01Tutorial (1).pdf, 2025

[2] OpenAI, ChatGPT, Loyola Marymount University, ChatGPT, 2025

ATTACHMENTS

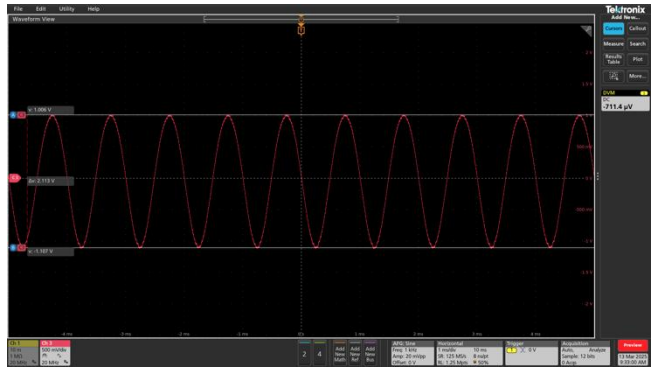


Figure 2: Vout of Overall Final Circuit

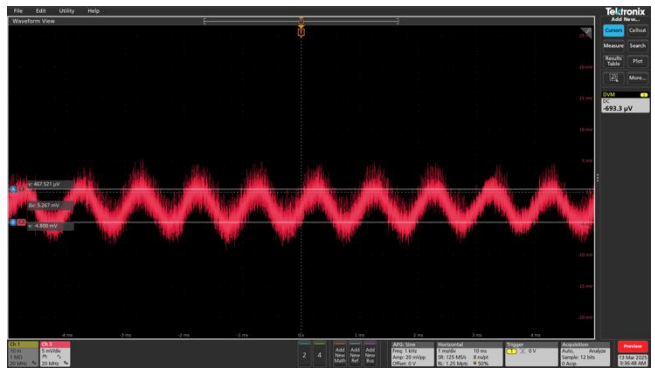


Figure 3: Vin of Overall Final Circuit

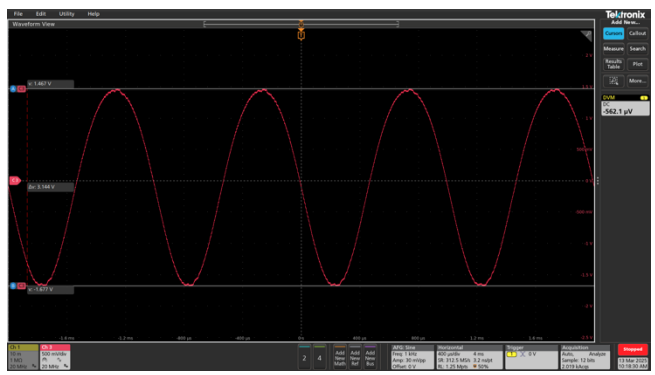


Figure 4: Vout at Distortion

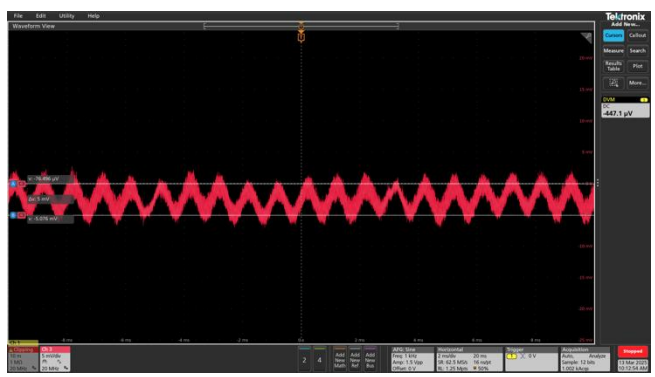
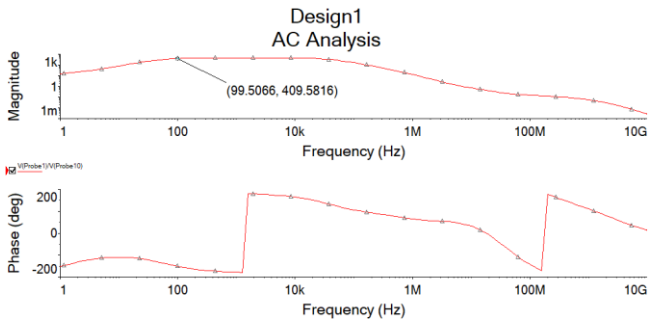
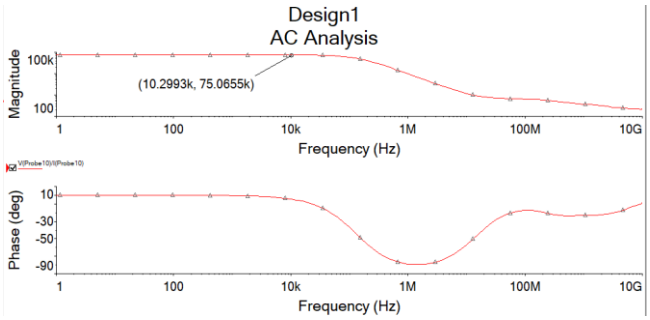


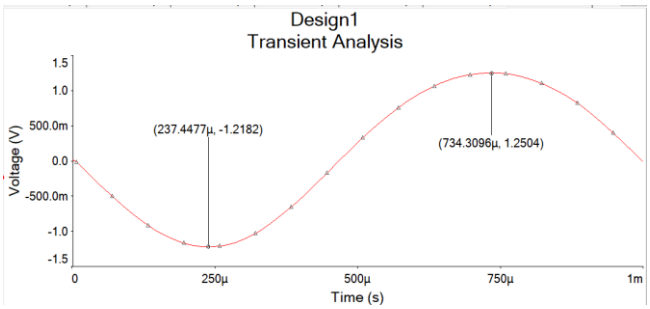
Figure 5: Vout in Common Mode



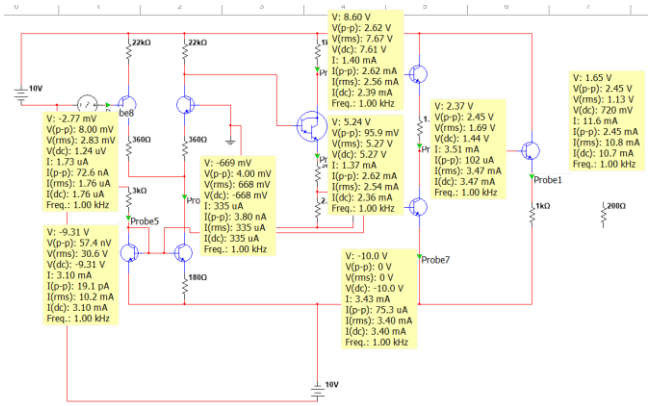
Voltage Gain Analysis



R_{id} Analysis



V_{out} Analysis



Verification of Circuit Components